Self-hosted WireGuard VPN for a home LAN

Slovak version: citajma.md

Reach your home network from a phone or laptop through a WireGuard tunnel that runs on your own NAS (Debian 12). No third-party coordination server: you only forward one UDP port on the router.

The project was built for a UniFi USG router (it has no VPN a phone can use), a Debian 12 NAS and a Samsung Galaxy S25+, but nothing in it is specific to those. The LAN and the DNS for clients are detected from the NAS network card (its subnet and default gateway). Defaults: tunnel `10.66.66.0/24` and `fd42:42:42::/64`, port `59656/udp`. All of them can be changed.

What is in the repository

Path	What it is
`scripts/regen-wg.sh`	Sets up the WireGuard **server** on the NAS (packages, `wg0.conf`, firewall rules, forwarding, service), then calls `add-wg-client.sh`.
`scripts/add-wg-client.sh`	Creates or regenerates one **client**: keys, PSK, config, QR code (PNG and in the terminal). Private files are shredded afterwards.
`webmin/`	A Webmin module (native Perl) that does the same from the browser, plus install detection, uninstall with cleanup and editing of the server settings. Available in 12 languages.
`docs/setup.en.md`	The full step-by-step guide (also in Slovak: `docs/setup.md`): router port forward, phone and Windows setup, troubleshooting, manual procedure.

The scripts print English only.

Quick start

On the NAS, as root:

```
sudo apt install wireguard iptables qrencode
chmod +x scripts/*.sh
sudo ./scripts/regen-wg.sh        # server + first device, shows a QR code
sudo ./scripts/add-wg-client.sh   # every further device
```

Then forward `UDP 59656` on the router to the NAS, scan the QR code in the WireGuard app and switch the tunnel on. The details (CGNAT check, DDNS, Android and Windows behaviour) are in `docs/setup.en.md`.

To manage the VPN from Webmin instead, see `webmin/readme.md`.

Security notes

- The scripts and the module run as root and put values into `wg0.conf` `PostUp`/`PostDown` lines, so the module validates every input strictly. Read the scripts before running them.

- Client private keys are shown once (QR code or file) and are not stored; the server keeps only the public keys.
- Never commit `wg0.conf`, client `.conf` files or keys. `.gitignore` blocks the common names, but check before you push.

Testing status

- The Webmin module is verified on a live Webmin 2.670 (Debian 12): installation, server setup, server overview, install status, adding a client including its actual connection, and deleting a client. Settings edit and uninstall were verified only against a simulated environment.
- The shell scripts are syntax-checked (`bash -n`). Try a full run on a test machine before relying on them on your NAS.

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